Results — Multi-Source IoT-23 Flow-Pattern Anomaly Detection

Madhav Dogra · 2026-06-30 · IoT-23 (6 scenarios, multi-source)

Setup

Data. Seven IoT-23 captures (1-1, 3-1, 20-1, 21-1, 42-1, 34-1, 8-1) → per-flow first K=32 packets × 13 payload-free features; labels joined from each Zeek `conn.log.labeled` by 5-tuple. Class-balanced and water-filled across captures into **31,960 flows**, in **3 classes**: Benign 10,000, Scan/Botnet 16,000, BruteForce 5,960. *Taxonomy:* the short-flow attack families — DDoS, Recon (port-scan) and Mirai C&C — are *inseparable in the first ∼6 packets* of payload-free data (see Takeaway), so they are reported as one **Scan/Botnet** class; IoT-23 `Attack` (telnet credential guessing) → BruteForce. Stratified 70/15/15 (test 4,794).

Model. Mamba encoder (d=128, 4 layers) + softmax classifier + one-class deep-SVDD anomaly head.

Run. 3 stages × 6 epochs, stratified split + $\sqrt{\cdot}$-inverse-frequency focal weights, CPU (Apple M3 Pro). **Training ≈1 h**; inference **5.0 ms/flow**. Stage B train acc →0.98; Stage C radius^2 0.0298 @ p99.

Results

Metric	Value
Accuracy	**98.2 %**
Macro-F1 (3 classes)	**0.984**
Weighted-F1	0.982
Anomaly ROC-AUC	**0.935**
Anomaly PR-AUC	0.959
Latency p50/p95/p99	5.0/6.5/6.9 ms

Confusion matrix (rows true, cols pred)

	B	S	F
Benign (B)	1433	63	4
Scan/Botnet (S)	16	2383	1
BruteForce (F)	0	1	893

Class	P	R	F1	N
Benign	0.989	0.955	0.972	1500
Scan/Botnet	0.974	0.993	0.983	2400
BruteForce	0.994	0.999	**0.997**	894
Macro avg	0.986	0.982	**0.984**	4794
Weighted avg	0.982	0.982	0.982	4794

Takeaway.

- **98.2 % accuracy**, macro-F1 **0.984**, over **7 captures** — BruteForce F1 0.997, Benign 0.972, Scan/Botnet 0.983.
- **Design finding:** payload-free header features on these short flows separate by flow *structure* — benign (∼3 pkts), short scan/botnet attacks (∼5 pkts), long brute-force (∼30 pkts) — *not* by fine attack family. DDoS, Recon and Mirai-C&C overlap in the first few packets (confirmed: class reweighting see-saws, and >99 % of these flows never reach 32 packets), so they form one Scan/Botnet class.
- Anomaly head ROC-AUC 0.935 on multi-source data (honest; the prior single-capture 0.994 was inflated by a Mirai-dominated scenario).
- Residual error is the benign↔scan boundary (short benign connections vs. short scans).

Multi-source run (6 IoT-23 scenarios). The 3-class taxonomy reflects what payload-free per-packet features can resolve on short flows; separating the attack families further (DDoS vs. Recon vs. Mirai) would require flow-level metadata — duration, byte counts, destination-port diversity, connection state.